A 1.5 V, 50 mA PMOS Low-Dropout Regulator in 0.18 μm CMOS

Part 2: Hand calculation of W/L from the specification, and ngspice verification

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Abstract

This note sizes a PMOS low-dropout regulator from a written specification and checks it in ngspice. Section 2 extracts square-law parameters from the 0.18 μm BSIM3 model. Section 3 computes every W/L, bias current and resistor step by step, following the system-level LDO design procedure of Rincón-Mora [1, 2]: topology, feedback network, pass device, open-loop gain, transient budget, gain stage, buffer and quiescent current. Section 4 verifies the design, reports four iterations, explains each gap between hand calculation and simulation, and lists the corner results. The final design meets every specification at the nominal point. **All equation, lemma, proposition and corollary numbers refer to Part 1** (`ldo_theory_proofs.pdf`), where each formula is proved; tables, figures and sections refer to this note.

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1 Specification and architecture

Table 1: Specification. Typical model, $T = 27\,^{\circ}\mathrm{C}$ unless stated. The last column is the simulated value of the final design (v4, Section 4).

Quantity	Condition	Spec	Simulated (v4)
Input voltage V_{IN}		1.7–1.9 V (nom. 1.8 V)	–
Output voltage V_{OUT}		1.50 V	1.4954–1.5053 V
Load current I_L		0.1–50 mA	–
Dropout voltage	$I_L = 50$ mA, V_{OUT} within 1 %	≤ 200 mV	141 mV
Output capacitor	ceramic	$C_L = 2.2\,\mu$F, $R_{ESR} = 0$–$50\,\Omega$	–
Phase margin	all loads, all R_{ESR} in range	$\geq 45^{\circ}$	57.0° (worst, incl. corners)
PSRR	1 kHz, $I_L = 50$ mA, $V_{IN} = 1.8$ V	≥ 30 dB	31.0 dB
Load regulation	I_L: 0.1$\rightarrow$50 mA	$\lvert\Delta V_{OUT}\rvert \leq 15$ mV	9.9 mV
Line regulation	V_{IN}: 1.7$\rightarrow$1.9 V	≤ 35 mV/V	29.9 mV/V (worst)
Quiescent current	excl. I_L, incl. divider	$\leq 100\,\mu$A	74.9 μA
Load transient	0.1$\leftrightarrow$50 mA, 1 μs edges	$\lvert\Delta V_{OUT}\rvert \leq 50$ mV	9.9 mV

Process and tools. The transistor model is the 0.18 μm BSIM3v3.1 file `cmos180.lib` (1.8 V devices, $t_{ox} = 4.1$ nm) from the analog-IC coursework. Simulations use ngspice-46 in batch mode, driven by Python scripts; every number in this note is read from the script outputs. The reference voltage $V_{REF} = 0.6$ V and the bias current $I_{REF} = 5\,\mu$A are ideal sources (a bandgap is outside the scope of this note).

Architecture (Fig. 1).

- *Pass device* M_P: a PMOS. A PMOS can work with $V_{IN} - V_{OUT}$ down to its saturation voltage, which is what makes a low dropout possible.
- *Feedback divider* R_1, R_2: it feeds back the fraction $\beta = R_2/(R_1 + R_2)$ of V_{OUT}.
- *Error amplifier* (EA) M_1–M_5: a five-transistor OTA with a PMOS input pair, which works at $V_{REF} = 0.6$ V. An NMOS pair would need its input common mode above $V_{GS} + V_{DS,sat} \approx 0.75$ V.
- *Buffer* M_B: a PMOS source follower with its bulk tied to its source, biased by M_{BS}. It drives the large gate capacitance of M_P from a low impedance (Part 1, Section 4.2).
- *Load*: the off-chip capacitor C_L with its series resistance R_{ESR}. The load current is an ideal current sink.

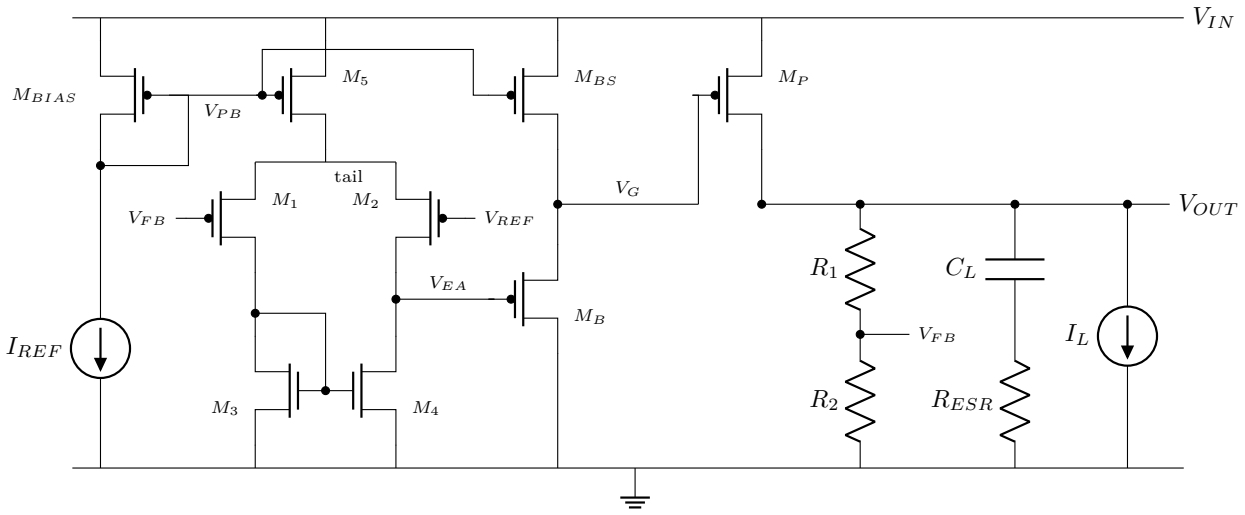


Figure 1: LDO schematic. M_1 (gate V_{FB}) sits on the diode side of the mirror, M_2 (gate V_{REF}) on the output side; Lemma 4 shows that this gives negative feedback. Bulk of every PMOS is V_{IN} except M_1, M_2 (tail node) and M_B (its source).

2 Device parameters

Extraction. For each device type and length, ngspice sweeps I_D–V_{GS} at $|V_{DS}| = 0.9$ V ($W = 10\,\mu$m). By (1), $\sqrt{I_D}$ is a straight line in V_{GS}. Its slope gives K' and its intercept gives V_{TH}; the fit uses $V_{OV} = 0.1$–0.3 V. Next, λ is the slope of I_D–V_{DS} at $V_{OV} = 0.2$ V over $|V_{DS}| = 0.6$–1.2 V, divided by I_D. Table 2 lists the results, and Fig. 2 shows the simulated g_m/I_D.

Table 2: Square-law parameters extracted from `cmos180.lib` (script `char_devices.py`).

	$L = 0.18\,\mu$m	0.5	1	2	3
NMOS V_{TH} (V)	0.479	0.459	0.428	0.410	0.404
NMOS K' (μA/V^2)	219	218	217	217	216
NMOS λ (V^{-1})	0.292	0.101	0.080	0.059	0.045
PMOS $\lvert V_{TH}\rvert$ (V)	0.465	0.431	0.412	0.399	0.394
PMOS K' (μA/V^2)	71.4	53.9	49.9	48.0	47.4
PMOS λ (V^{-1})	0.342	0.118	0.077	0.046	0.032

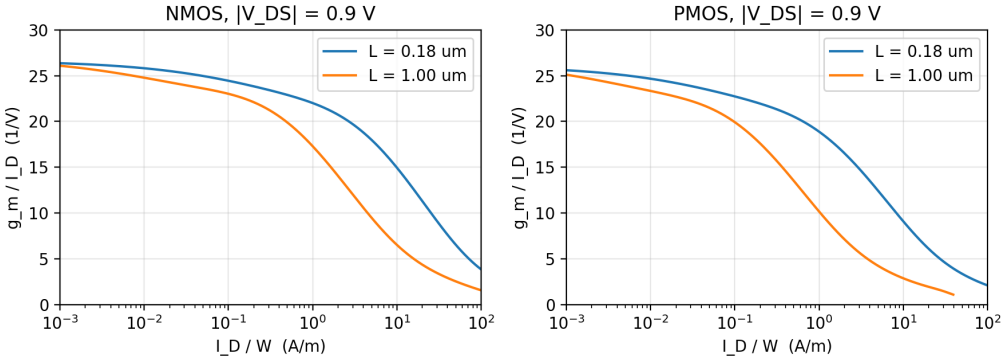


Figure 2: Simulated g_m/I_D versus current density. The square law predicts $g_m/I_D = 2/V_{OV}$, which is $20\,\mathrm{V}^{-1}$ at $V_{OV} = 0.1\,\mathrm{V}$. The real device gives about $17\,\mathrm{V}^{-1}$ there (moderate inversion) and levels off near $1/(nV_T) \approx 26\,\mathrm{V}^{-1}$ in weak inversion; this is one source of the hand-versus-simulation gap in Section 4.3.

Note on the model file. ngspice reports that the model parameters `XL` and `XW` are not recognised and ignores them. In this file $XL = 0$ and $XW = -10\,\mathrm{nm}$, so the effect is a 10 nm width change and is negligible for the widths used here.

3 Hand design: the LDO design procedure

The sizing follows the system-level design procedure of Rincón-Mora: the design considerations of [1, Chapter II] (AC analysis, transient analysis, pass device, amplifier), the pass-element choice of [1, Section 6.1], and the dropout, buffer and gain-stage equations of [2, Sections II–IV]. Each formula is used in the form proved in Part 1. The script `design_flow.py` carries out the eight steps below, and each number is quoted from its output. Steps 6 and 7 run twice. *Pass A* uses square-law parameters only. *Pass B* replaces g_m/I_D and λ by the values read at the simulated operating point of the previous design (v3, Section 4.1), the usual second pass of a hand design. The result is the final design v4.

Step 1 – Topology. Of the five pass elements compared in [1, Table 6.1], the common-source PMOS has the lowest dropout ($V_{SD,sat}$) together with a low quiescent current, and the reference calls it the best overall choice, so M_P is a PMOS. Its large gate capacitance must not load the high-impedance gain stage, so a buffer isolates the two (the buffered architecture of [1, Fig. 2.7]). The gain stage is the five-transistor OTA of [2, Fig. 8(a)], which needs only a small common-mode range around V_{REF} [2, Section IV]. With $V_{REF} = 0.6\,\mathrm{V}$ a PMOS input pair is used (Step 6a). The buffer is a class-A PMOS source follower M_B biased by M_{BS}, with its bulk tied to its source.

Step 2 – Feedback network [1, Section 2.1]. The divider current is a quiescent-current overhead, so it is kept small: $I_R = 10\,\mu\mathrm{A}$. Then $\beta = V_{REF}/V_{OUT} = 0.40$, $R_2 = V_{REF}/I_R = 60\,\mathrm{k\Omega}$ and $R_1 = (V_{OUT} - V_{REF})/I_R = 90\,\mathrm{k\Omega}$.

Step 3 – Pass device [2, Eqs. (12)–(14)], [1, Section 2.3]. Two conditions bound W/L from below.

- *Dropout.* The buffer can pull V_G down to $V_{G,min} = V_{OV4} + V_{SG,B} \approx 0.80\,\mathrm{V}$ ($V_{SG,B}$ is checked in Step 6b). At $V_{IN,min}$ the available drive is $V_{SG,P} = 0.899\,\mathrm{V}$, i.e. $V_{OV} = 0.434\,\mathrm{V}$. Carrying I_{max} at $V_{SD} = V_{DO}$ (triode, (7), the exact form of [2, Eq. (12)]) needs $W/L \geq 10465$.
- *Saturation at $V_{IN,nom}$.* The loop gain and the PSRR collapse in triode (Corollary 20), so M_P must stay saturated at the nominal input. By (6) with a 10 mV margin, $W/L \geq 16644$.

The saturation condition is binding. Rounding up to fingers of $10\,\mu\mathrm{m}$ gives $M_P = 300 \times 10\,\mu\mathrm{m}/0.18\,\mu\mathrm{m}$ ($W/L = 16667$). With this size the dropout point needs only $V_{SG,P}^{DO} = 0.775\,\mathrm{V}$, below the available 0.899 V, and the on-resistance at the available drive is $R_{on} = 1/(K'\frac{W}{L}V_{OV}) = 1.93\,\Omega$ [2, Eq. (13)]. At full load $V_{SG,P} = 0.755\,\mathrm{V}$ and $g_{m,P} = 0.345\,\mathrm{S}$. At light load (weak inversion) the simulated I_D–V_{SG} curve of one finger gives $V_{SG,P} = 0.366\,\mathrm{V}$ and $g_{m,P} = 2.38\,\mathrm{mS}$, so $\Delta V_{SG,P} = 389\,\mathrm{mV}$. The gate capacitance, C_{par} in [1, 2], is $C_G = \frac{2}{3}WLC_{ox} + 2WC_{ov} = 6.96\,\mathrm{pF}$.

Step 4 – Open-loop gain from regulation and PSRR [1, Eq. (2.9)]. Equation (2.9) of [1] gives the closed-loop output resistance $R_{o,pass}/(1+A_{ol}\beta) \approx 1/(Gg_{m,P})$ with $G = \beta A_{EA} A_B$. Since $g_{m,P}$ changes by two decades over the load range, it is integrated over the range, which is Corollary 6(b): $|\Delta V_{OUT}| = \Delta V_{SG,P}/G$. Each specification gives a lower bound on G: PSRR (Proposition 19): $G \geq 31.6$; load regulation: $G \geq 25.9$; line regulation (Corollary 6c): $G \geq 28.6$. The PSRR is binding: $G_{min} = 31.6$.

Step 5 – Transient budget [1, Eqs. (2.10)–(2.12)]. By Proposition 23, the 50 mV limit for the $\Delta I = 49.9\,\text{mA}$ step allows a response time

$$\Delta t_{1,max} = \frac{C_L\,(\Delta V_{tr} - \Delta I\, R_{ESR})}{\Delta I} = 2.18\,\mu\text{s}.$$

The loop reaction time $1/(2\pi f_u)$ and the gate slew time (20) must fit in this budget. Steps 6 and 7 check both.

Step 6 – Gain stage [1, Section 2.4], [2, Section IV].

(a) *Common-mode range around V_{REF}.* For a PMOS pair with an NMOS mirror, $V_{IC,max} = V_{IN,min} - V_{SD,sat5} - V_{SG1} = 1.038\,\text{V}$ and $V_{IC,min} = V_{GS3} - |V_{TH1}| = 0.142\,\text{V}$. $V_{REF} = 0.6\,\text{V}$ lies inside.

(b) *Output swing.* The gain stage must move the buffer over the whole range from M_P almost off to full drive [2, Section IV]. For this topology that is the window of Proposition 8: $0.522\,\text{V} \leq V_{SG,B} \leq 0.775\,\text{V}$. Its centre, $V_{SG,B} = 0.65\,\text{V}$, sets $V_{OV,B} = 0.22\,\text{V}$.

(c) *Gain and channel lengths.* With $V_{OV1} = 0.1\,\text{V}$, $A_{EA} = (2/V_{OV1})/(\lambda_p + \lambda_n)$ (Lemma 10). By Proposition 16, $A_{EA}\,p_{EA}$ is fixed for a given L_{12} and falls as L_{12} grows, so any gain above the need costs bandwidth. The rule is therefore: the shortest L_{12}, then the shortest L_{34}, whose square-law G exceeds G_{min} by 6 dB. The margin covers the known optimism of the square law (g_m/I_D too high, λ too low; Fig. 2, Section 4.3), which iterations v1 and v2 measured. Table 3 lists the candidates; the rule selects $L_{12} = 1\,\mu\text{m}$, $L_{34} = 3\,\mu\text{m}$, with $A_{EA} = 164$ and $G = 64.1$.

(d) *Current.* By Proposition 16, p_{EA} does not depend on I_{D1}, so the bandwidth requirement of [1, Section 2.4] does not fix it; $I_{D1} = 20\,\mu\text{A}$ per side is a budget choice. Then $p_{EA} = 2.02\,\text{MHz}$ (Pass A).

Table 3: Step 6c, square-law gain for each pair of channel lengths ($V_{OV1} = 0.1\,\text{V}$, $A_B = 0.979$). The first row that meets $G \geq 2G_{min} = 63.2$ in the order (L_{12}, L_{34}) is chosen.

L_{12} (μm)	L_{34} (μm)	A_{EA}	G	G/G_{min}	meets rule
1	1	127	49.8	1.58	no
1	2	147	57.7	1.82	no
1	3	164	64.1	2.03	yes
2	1	159	62.2	1.97	no
2	2	192	75.0	2.37	yes
2	3	220	86.1	2.72	yes
3	1	177	69.5	2.20	yes
3	2	219	85.8	2.71	yes
3	3	257	100.7	3.19	yes

Step 7 – Buffer [2, Eq. (3)], [1, Section 2.4]. The parasitic pole at the gate of M_P, $p_B = g_{out}/(2\pi C_G)$ (Lemma 11; P_3 in [2]), must lie above f_u. The worst case is full load [1, Section 2.1] with $R_{ESR} \to 0$. Proposition 15 with a target of 60° (the specification, 45°, plus 15° for corners) gives $\arctan(f_u/p_B) \leq 30° - \arctan(f_u/p_{EA})$.

- *Pass A.* $f_u = 1.60\,\text{MHz}$ and $\arctan(f_u/p_{EA}) = 38.3°$ leave −8.3° for the buffer: no bias current reaches the target. The square-law gain, and with it f_u, is too high, which is exactly the optimism the 6 dB margin of Step 6c anticipated.
- *Pass B.* At the operating point of v3: $g_{m1}/I_{D1} = 16.6\,\text{V}^{-1}$, $\lambda_2 = 0.102$, $\lambda_4 = 0.069\,\text{V}^{-1}$, so $A_{EA} = 97.4$ and $G = 38.2$ (still $1.21 \times G_{min}$). Then $f_u = 893\,\text{kHz}$, $p_{EA} = 2.83\,\text{MHz}$, $\arctan(f_u/p_{EA}) = 17.5°$, and the buffer pole must be at least $p_B = 4.03\,\text{MHz}$: $g_{mB} \geq 2\pi C_G p_B A_B = 0.173\,\text{mS}$. With the operating-point $g_{mB}/I_B = 9.47\,\text{V}^{-1}$ this needs $I_B \geq 18.2\,\mu\text{A}$.
- *Slew current.* For the load release only I_B charges C_G (Remark after Proposition 23). With $1/(2\pi f_u) = 0.178\,\mu\text{s}$, the budget of Step 5 leaves $t_{sr} \leq 2.00\,\mu\text{s}$, so (20) needs $I_B \geq 1.35\,\mu\text{A}$. The pole condition is binding.

$I_B = 20\,\mu$A (the next multiple of I_{REF}; v3 used 30 μA). Predictions: $p_B = 4.42$ MHz, PM = 61.1° ($R_{ESR} = 0$) and 68.1° ($R_{ESR} = 10\,\text{m}\Omega$), $t_{sr} = 0.14\,\mu$s. The load-step excursion by (19) is at most 29 mV with the conservative $t_{loop} = 1/f_u$ of [1], or 7.6 mV with $t_{loop} = 1/(2\pi f_u)$, both within 50 mV. M_{BS} mirrors I_{REF} ($W_{BS} = 20\,\mu$m) and M_B is sized by (1) at $V_{OV,B}$: $W_B = 8\,\mu$m, $L_B = 0.5\,\mu$m.

Step 8 – Quiescent current and current efficiency [2, Eq. (1)]. $I_Q = I_{REF} + 2I_{D1} + I_B + I_R = 75\,\mu$A (v3: 85 μA), within the 100 μA limit. The current efficiency $I_L/(I_L + I_Q)$ is 57 % at 0.1 mA and 99.85 % at 50 mA. Pass B removes the 10 μA by which the buffer of v3 was over-biased, which is the aim of the procedure: the least quiescent current that the stability and transient requirements allow [2].

Table 4: Final device sizes (v4). M_{BIAS}, M_5 and M_{BS} form a mirror; the widths set the current ratios. Only M_{BS} and M_B differ from v3.

Device	W (μm)	L (μm)	Role / current / set by
M_{BIAS}	5	1	diode, $I_{REF} = 5\,\mu$A
M_5	40	1	tail, 40 μA (Step 6d)
M_1, M_2	81	1	input pair, $V_{OV} \approx 0.1$ V (Steps 6a, 6c)
M_3, M_4	25	3	mirror load, $V_{OV} \approx 0.15$ V (Step 6c)
M_{BS}	20	1	buffer bias, 20 μA (Step 7)
M_B	8	0.5	source follower, bulk = source (Steps 6b, 7)
M_P	300×10	0.18	pass device (Step 3)
R_1 / R_2	90 kΩ / 60 kΩ		divider (Step 2)

4 Simulation

Method. The loop gain uses the break of Proposition 9. A 10^9 H inductor from the divider tap to the gate of M_1 closes the loop at DC, and a 10^3 F capacitor injects the AC test voltage at the gate. Because the gate draws no current, this measures (9) exactly for $f \gg 1/(2\pi\sqrt{LC})$. The PSRR test puts an AC source on V_{IN} with the loop closed. Line, load and dropout are DC sweeps, and the load step is a transient run with 1 μs edges.

Static output equation. Insert the simulated operating-point voltages and the simulated $A_{EA} = g_{m1}/(g_{ds2} + g_{ds4})$ into (5):

- Light load: $\varepsilon = 214$ mV, so the formula gives $V_{OUT} = 1.5055$ V. Simulation: 1.5053 V.
- Heavy load: $\varepsilon =$-183 mV, so the formula gives 1.4953 V. Simulation: 1.4954 V.

The equation therefore explains both the ±5 mV output error and its change of sign with load.

4.1 Design iterations

Table 5: Four iterations (nominal: $V_{IN} = 1.8$ V, 27 °C, $R_{ESR} = 10\,\text{m}\Omega$; corner columns: $V_{IN} \in \{1.8, 1.9\}$ V, $T \in \{0, 27, 85\}$ °C, both load ends). Values that miss the specification are marked with ×.

	v1	v2	v3	v4 (final)
L_{12} / L_{34} (μm), $V_{OV,B}$ (V)	1 / 1, 0.15	2 / 2, 0.15	1 / 3, 0.22	1 / 3, 0.22
Buffer current I_B (μA)	30	30	30	20
Simulated A_{EA} at full load	76	119	99	99
PSRR at 1 kHz, 50 mA (dB)	28.8 ×	32.7	31.0	31.0
Line regulation, worst (mV/V)	39.1 ×	28.5	29.9	29.9
PM at 50 mA (deg)	81.4	55.8	71.1	68.3
PM, $R_{ESR} = 0$, worst corner (deg)	–	44.4 ×	60.0	57.0
PSRR at 1 kHz, worst corner (dB)	–	22.2 ×	29.8	29.8
I_Q (μA)	84.7	84.7	84.7	74.9

- **v1** ($L = 1\,\mu$m everywhere) is stable, with a large margin, but its EA gain is only 76 against 127 by the square law. As a result, PSRR and line regulation miss. This is the gain optimism that the 6 dB margin of Step 6c now covers.

- **v2** doubles L for M_1–M_4. The gain rises, but by Proposition 16 p_{EA} falls (it scales as λ/L), and the phase margin with $R_{ESR} = 0$ drops to 44.4° at $V_{IN} = 1.9\,\mathrm{V}$, 85 °C. At light load and $V_{IN} = 1.9\,\mathrm{V}$, V_{EA} also rises to within 0.07 V of the bound $V_{REF} + |V_{TH,p}|$ of Proposition 8. M_2 then nears triode and the PSRR falls to 22.2 dB. This is the reason for the shortest-L_{12} rule of Step 6c.
- **v3** lengthens only the mirror ($L_{34} = 3\,\mu\mathrm{m}$), because the simulated operating point shows that g_{ds4}, not g_{ds2}, dominates the EA output conductance, and moves $V_{OV,B}$ to the centre of the window (8). Its buffer current, 30 μA, was a budget choice. v3 meets every specification.
- **v4** is the outcome of the procedure of Section 3: Step 6c reproduces the lengths of v3, and Pass B of Step 7 sets the buffer current from the pole and slew conditions, 20 μA. The quiescent current falls by 9.8 μA; the phase margin at full load falls by 2.8° and stays above the 60° target with $R_{ESR} = 0$ at the nominal point.

4.2 Results of the final design

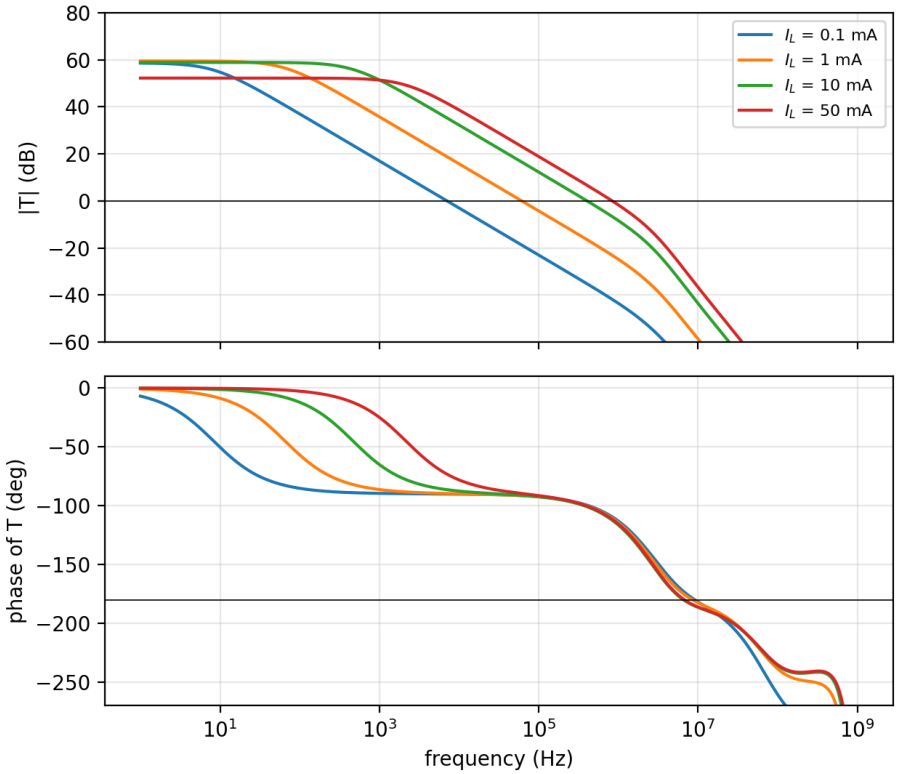


Figure 3: Loop gain T at four loads. The DC gain is about 52–60 dB, f_u moves from 7.1 kHz to 840 kHz with load (Proposition 14), and the PM is 68.3° at the worst (full) load.

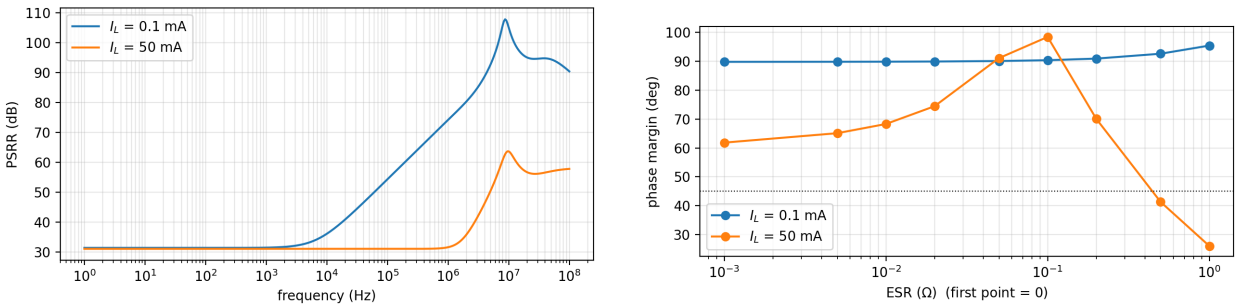


Figure 4: Left: PSRR. It is flat at about G up to the loop bandwidth, as predicted by (17); above f_u the capacitor C_L takes over. Right: phase margin versus R_{ESR}. The margin first rises (zero lead), peaks near 100 mΩ, then collapses as the plateau of Proposition 17 pushes f_u toward p_B, p_{EA}. It stays above 45° up to 200 mΩ and is 41° at 500 mΩ.

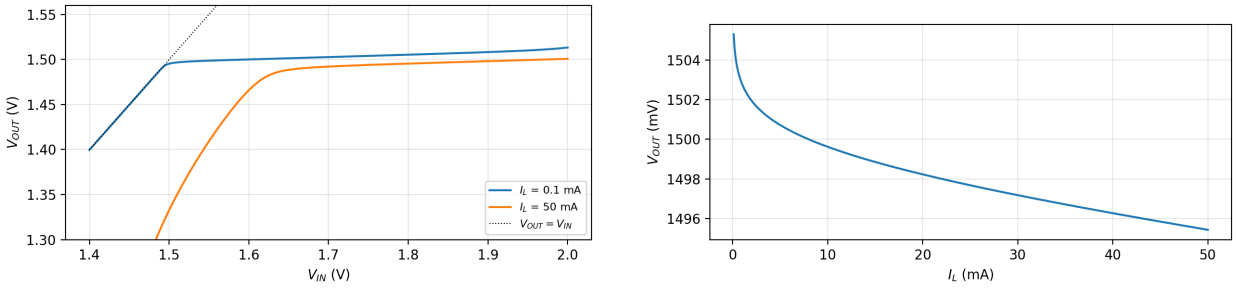


Figure 5: Left: V_{OUT} versus V_{IN}. The dropout (1 % criterion) at 50 mA is 141 mV. Right: load regulation, $\Delta V_{OUT} = 9.9$ mV over the full range.

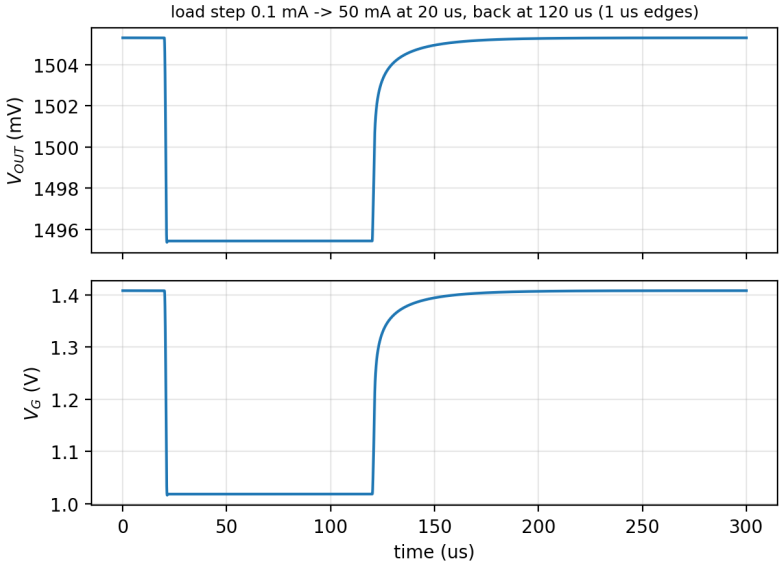


Figure 6: Load step. The step-up settles within 1 mV immediately (fast loop at full load). The release settles exponentially with a fitted $\tau = 20.9\,\mu$s, against $1/(2\pi f_{u,light}) = 22.4\,\mu$s from Proposition 21. There is no overshoot, as the single-pole model predicts. The total excursion, 9.9 mV, is almost all the static load regulation (9.9 mV).

4.3 Hand calculation versus simulation

Table 6: Final design at full load: the two passes of Section 3 and the simulation of v4. Pass A uses square-law parameters; Pass B uses g_m/I_D and λ read at the v3 operating point.

Quantity (full load)	Pass A	Pass B	Simulated (v4)
g_{m1}/I_{D1} (V^{-1})	20	16.6	16.6
λ_{eff} of M_2 / M_4 (V^{-1})	0.077 / 0.045	0.102 / 0.069	0.102 / 0.069
A_{EA}	164	97	99
$G = \mathrm{PSRR}_{DC}$ (dB)	36.1	31.6	31.0
f_u (kHz)	1599	893	840
PM, $R_{ESR} = 10\,\mathrm{m\Omega}$ (deg)	–	68.1	68.3
PM, $R_{ESR} = 0$ (deg)	–	61.1	61.9
Load regulation (mV)	6.1	10.2	9.9
I_Q (μA)	–	75	74.9

Pass A misses in two square-law inputs; the formulas themselves hold.

(i) g_m/I_D. The input pair works at $V_{OV} \approx 0.08$ V, in moderate inversion, where g_m/I_D saturates toward $1/(nV_T)$ (Fig. 2) instead of following $2/V_{OV}$. The simulated value is 16.6 against 20, about 1.6 dB of gain.

(ii) λ. The extraction was done at $V_{OV} = 0.2$ V and $|V_{DS}| = 0.6$–1.2 V. In the circuit, M_2 has a smaller

V_{OV} and M_4 sits at $V_{DS} = 0.37\,\mathrm{V}$, close to saturation, where g_{ds} is larger. The effective λ values are higher than the table values.

Pass B puts the operating-point values into the same formulas and predicts the phase margin of v4 to within 1°, the loop gain to within 1 dB, f_u to within 6 % and the quiescent current to within 1 μA. Because Pass A overestimates G, it overestimates f_u and finds no buffer current that reaches the PM target; the second pass is what makes the buffer sizing usable.

The same mechanism affects the pass device. At full load $V_{SD} = 0.30\,\mathrm{V}$ is close to $V_{DS,sat} = 0.22\,\mathrm{V}$, so $g_{ds,P} = 30\,\mathrm{mS}$ against $\lambda I = 17\,\mathrm{mS}$. This lowers R_o and the DC loop gain (52.3 dB simulated against 62.2 dB by the square law), but it does not move f_u, which is independent of R_o (Proposition 14).

4.4 Corners

The model file has no process corners, so the checks cover V_{IN} and temperature only (script `sim_extra.py`).

- *Stability.* For $V_{IN} \in \{1.7, 1.8, 1.9\}\,\mathrm{V}$, $T \in \{0, 27, 85\}\,^\circ\mathrm{C}$ and both load ends, the minimum PM is 63.2° at $R_{ESR} = 10\,\mathrm{m\Omega}$, and 57.0° with $R_{ESR} = 0$ (at 1.9 V, 85 °C).
- *Output voltage.* V_{OUT} stays within 1.4876–1.5088 V over all corners.
- *PSRR, $V_{IN} \geq 1.8$ V.* The worst value is 29.8 dB at 1.9 V, 0 °C, 0.1 mA. This is just below the 30 dB specification, which is stated at the nominal point; the cause is the remaining headroom limit of Proposition 8 at high V_{IN}.
- *PSRR, $V_{IN} = 1.7$ V.* At full load this is the dropout region, and the PSRR falls to 15.6 dB at 85 °C. Corollary 20 explains it: at the 27 °C dropout point, the simulated ratio $g_{ds,P}/g_{m,P}$ is already 0.33 (against 0.09 at 1.8 V), and it grows further when hot. Datasheets specify PSRR with V_{IN} well above $V_{OUT} + V_{DO}$ for this reason.

5 Limitations and next steps

- V_{REF} and I_{REF} are ideal. A bandgap reference and its PSRR would add to the supply path.
- Only the typical model is available: no process corners and no mismatch (Monte Carlo) analysis. Random offset of the input pair adds to ε in (5).
- The design is schematic-level only. Layout parasitics, bond-wire inductance in series with C_L, and the pass-device metal resistance are not included.
- Not covered: start-up, current limit and short-circuit protection, noise, and the soft-start of V_{REF}.
- The PSRR is limited to about G by the ground-referenced gate (Corollary 18). Possible improvements are a second EA gain stage with its own compensation, or a gate drive that tracks V_{IN}.

A Reproducing the results

The scripts, results and document sources are in the GitHub repository `https://github.com/NguyenVuQuocMinh/cmos-ldo-regulator-180nm`. The 0.18 μm model file is not redistributed there; place a BSIM3 model with `.model NMOS` and `.model PMOS` as `sim/cmos180.lib`, and set the environment variable `NGSPICE` to the ngspice console executable if it is not at the default path. Run from `scripts/`:

1. `python char_devices.py`: device parameters (Table 2, Fig. 2).
2. `python design.py`: earlier iterations, writes `sizes_v1/v2/v3.json` and `hand.json`.
3. For v1–v3: set `LDO_SIZES=sizes_vN.json` and `LDO_VER=vN`, then run `sim_op.py`, `sim_all.py` and `sim_extra.py`.
4. `python design_flow.py`: the procedure of Section 3 (Pass B reads the v3 operating point), writes `sizes_v4.json` and `hand_flow.json`. Then run the three simulation scripts for v4.
5. `python render.py <path-to-tectonic.exe>`: fills this document with the numbers and builds the PDF.

The netlist is generated by `ldo_netlist.py` from the sizes file. The repository folder `netlists/` holds the final circuit (v4) with its operating-point, loop-gain, PSRR and load-step test benches; each runs directly with `ngspice_con -b` once the model file is in place.

References

[1] G. A. Rincón-Mora, *Current Efficient, Low Voltage, Low Drop-Out Regulators*, Ph.D. dissertation, Georgia Institute of Technology, 1996, Chapter II and Section 6.1. Author's copy: `https://rincon-mora.gatech.edu/publicat/books/thesis/ldo_book.pdf`.

[2] G. A. Rincón-Mora and P. E. Allen, "A low-voltage, low quiescent current, low drop-out regulator," *IEEE J. Solid-State Circuits*, vol. 33, no. 1, pp. 36–44, Jan. 1998. Author's copy: `https://rincon-mora.gatech.edu/publicat/jrnls/jssc98_lv_ldo.pdf`.